YUXIANG MA

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ABOUT ME

PhD candidate at MIT working on visuo-tactile perception and robot learning for contact-rich manipulation. My work runs from tactile sensor and gripper design, through human and robot data collection, to video world models and foundation-model systems deployed on real robots.

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA *Sep. 2021–Present*
PhD and S.M. (2023), Mechanical Engineering. Advisor: Edward Adelson. GPA: 5.0/5.0

Tsinghua University, Beijing *Aug. 2016–Jul. 2020*
B.Eng., Engineering Mechanics, Tsien Excellence in Engineering Program. GPA: 3.85/4.0, rank 2/28

WORK EXPERIENCE

Research Intern, Robotics and AI Institute, Cambridge, MA *Jun. 2025–Aug. 2025*
- Built a handheld UMI-style data collection device with force-torque and tactile sensors (Team Capture).
- Wrote the synchronized software that records visuo-tactile manipulation data for robot learning.

RESEARCH

CoStream: Composing Simple Behaviors for Complex Manipulation *2025–2026*
GelSight Lab, MIT & Embodied Minds Lab, Harvard *CoRL 2026, project lead*
- Split manipulation into independent behaviors: spatial constraints from LLMs/VLMs, trajectories from a video world model, and high-rate tactile and force corrections.
- Each behavior outputs an SE(3) target, and the targets compose into one pose command per control step.
- Solves 8 real-world tasks, including CPU/GPU/RAM insertion, without per-task demos or retraining.

ACTS: Visuo-Tactile World Model Learned from Human Physical Interaction *2026*
GelSight Lab, MIT *arXiv 2026, project lead*
- Built React, a 4.9-hour bimanual human dataset from handheld GelSight sensors (scene, wrist, and tactile video, 6-DoF motion, normal force).
- Fine-tuned Cosmos with LoRA to jointly predict five visual and tactile streams from force-aware actions.
- 0.7–2.0 dB scene PSNR gain over visuo-tactile baselines; zero-shot tactile transfer to robot teleoperation.

Tactile 6-DoF Object Pose Estimation (YOTO, InvariantCloud) *2025–2026*
GelSight Lab, MIT, with HKUST and University of Cambridge *ICRA 2026 and preprint, project lead*
- YOTO recovers 6-DoF object pose from one pair of tactile contacts on a CAD or phone-scanned mesh.
- A network pretrained on virtual tactile patches localizes contacts; closed-form SVD returns the pose.
- Stays accurate under occlusion, where FoundationPose drifts.
- InvariantCloud tracks pose from globally invariant marker point clouds, reducing yaw drift.

Tactile Sensors and Fingers (GlowTact, GelLink, GelSight Fin Ray) *2021–2026*
GelSight Lab, MIT *ICRA 2024, RoboSoft 2024, and preprint; project lead*
- GlowTact: a compact sensor using only single-color light and a monochrome camera, with no photometric stereo or calibration; detects a 0.2 g screw and resolves 0.25 mm threads.
- GelLink: a three-phalanx, single-motor finger that uses one camera for both touch and proprioception.
- Fin Ray: FEM and rendering simulation for designing compliant tactile fingers.

SKILLS

Programming: Python, C/C++, ROS
Learning: video world models, tactile representation learning, LoRA fine-tuning
Hardware/simulation: SolidWorks, Onshape, ABAQUS, Mitsuba, Blender, Unity, machine shop

PUBLICATIONS (* equal contribution)

2026

CoRL CoStream: Composing Simple Behaviors for Generalizable Complex Manipulation
H. Chen*, **Y. Ma***, S. Tian, X. Han, W. Huang, F. Wu, Y. Li, J. Wu, E. Adelson, Y. Du [website]

ICRA InvariantCloud: A Globally Invariant, Uniquely Indexed Point Cloud Framework for Robust 6-DoF Tactile Pose Tracking
P. Ye*, **Y. Ma***, Y. Zhou, W. Chen, W. Dong, M. Duan [arXiv]

WAFR RichMap: A Reachability Map Balancing Precision, Efficiency, and Flexibility for Rich Robot Manipulation Tasks
Y. Lu, **Y. Ma**, J. Pan [arXiv] [code]

arXiv ACTS: Learning Visuo-Tactile World Models from Human Physical Interaction
Y. Ma*, W. Xu*, P. Ye, A. Mei, Y. R. Song, T. Li, H. Chen, E. Adelson [arXiv]

arXiv Robot Learning with Visual Predicted Force
H. Chen*, F. Wu*, **Y. Ma**, M. Mete, P. Ye, . . . , R. D. Howe, Y. Du [website]

arXiv GlowTact: Simple and Compact Vision-Based Tactile Sensing with High Sensitivity and Spatial Resolution
Y. Ma*, M. Tippur*, P. Ye, S. Q. Liu, H. Chen, F. R. Cottrell, E. Adelson [website]

arXiv A Brain-inspired Hierarchical Framework for Zero-Shot Robot Task Reasoning and Execution
G. Wang, P. Ye, Q. Ying, Y. Jing, **Y. Ma**, H. Chen, . . . , M. Duan, B. Sheil [arXiv]

arXiv RoboEdit: Turning Human Manipulation Videos into Scalable Robot Experience
Y. Guo, Z. Tao, Y. Jiang, Y. Chen, Z. Dou, **Y. Ma**, Y. Yang, . . . , C. Jiang [website]

arXiv You Only Touch Once: 6-DoF Object Pose Estimation from Single Tactile Contact
P. Ye*, **Y. Ma***, H. Chen, G. Wang, Y. Jing, B. Sheil, Y. Du, E. Adelson [arXiv]

2025

ICRA-W Dynamic Regrasping with Asynchronous Vision Feedback using a Minimalist Robotic System
Y. Ma, E. Adelson *RoboLetics 2.0 Workshop* [paper]

2024

CoRL Transferable Tactile Transformers for Representation Learning Across Diverse Sensors and Tasks
J. Zhao, **Y. Ma**, L. Wang, E. Adelson [website]

ICRA GelLink: A Compact Multi-phalanx Finger with Vision-based Tactile Sensing and Proprioception
Y. Ma, J. Zhao, E. Adelson [arXiv]

RoboSoft Scalable, Simulation-Guided Compliant Tactile Finger Design
Y. Ma*, A. Agarwal*, S. Q. Liu*, W. Yuan, E. Adelson [arXiv]

2023

Nat. BME Stretchable Ultrasonic Arrays for the Three-Dimensional Mapping of the Modulus of Deep Tissue
H. Hu*, **Y. Ma***, X. Gao*, D. Song*, . . . , S. Xu *Nature Biomedical Engineering* [paper]

RoboSoft GelSight Baby Fin Ray: A Compact, Compliant, Flexible Finger with High-Resolution Tactile Sensing
S. Q. Liu, **Y. Ma**, E. Adelson [arXiv]

Nature A Wearable Cardiac Ultrasound Imager
H. Hu, H. Huang, M. Li, X. Gao, L. Yin, R. Qi, R. S. Wu, X. Chen, **Y. Ma**, . . . , S. Xu [paper]

CoRL-W Object Recognition and Force Estimation with the GelSight Baby Fin Ray
S. Q. Liu, **Y. Ma**, E. Adelson *Workshop on Learning for Soft Robots* [arXiv]

2021

EML The Effect of Arterial Stiffness on Cuff-Based Blood Pressure Measurement
Y. Ma, Y. Chen, Y. Ma, X. Feng *Extreme Mechanics Letters* [paper]